SC7 Bayes Methods MT25

Lecturer: Geoff Nicholls

Lecture 14: Bayesian non-parametrics, the Multinomial Dirichlet Process and the Dirichlet Process

Bayesian Non-parametrics.

If data $y_1, y_2, ...$ are an IES then by DeF,

$$p(y_1, ..., y_n) = \int_\Omega p(y_1, ..., y_n|\theta) dG(\theta).$$

These distributions and the posterior distribution

$$d\pi(\theta|y_1, ..., y_n) \propto p(y_1, ..., y_n|\theta) dG(\theta)$$

exist with G unknown. In Bayes, unknowns are random variables.

Say $G \in \mathcal{G}$ (space of possible true generative models).

Π is a distribution over $\mathcal{G}$ (ie, over distributions, like $\pi(\theta|M)$).

A prior $\pi \in \mathcal{G}$ is a realisation $G = \pi$ of $G \sim \Pi$.

We have a prior $d\Pi(G)$ for our prior!

We might *very informally* work with the joint distribution of G, θ,

$$d\Pi(G, \theta) = dG(\theta) d\Pi(G)$$

in which case the posterior would be

$$d\Pi(G, \theta | y) \propto p(y|\theta) dG(\theta) d\Pi(G).$$

We will work with the marginal prior

$$d\pi(\theta) \propto \int_{\mathcal{G}} dG(\theta) d\Pi(G).$$

The posterior $d\pi(\theta|y) \propto p(y|\theta) d\pi(\theta)$ is hiding alot!

Prior elicitation moves up a level in the hierarchy to the choice of Π, a distribution over distributions.

Mixture models (target application)

Data $y = (y_1, \ldots, y_n)$, $y_i \in \mathbb{R}$ come in K groups/clusters.

Set $S_k, k \in [K]$ of labels $i \in [n]$ gives samples in group k.
$S = (S_1, \ldots, S_K)$ is a partition of $[n]$.

Observation model in cluster k is $y_i \sim f(\cdot|\theta_k^*)$, likelihood is

$$f(y|\theta^*, S) = \prod_{k=1}^{K} \prod_{i \in S_k} f(y_i|\theta_k^*).$$

Clustering: K, S and $\theta^* = (\theta_1^*, \ldots, \theta_K^*)$ estimated jointly.

Priors $\pi(\theta^*|S) = \prod_{k=1}^{K} \pi(\theta_k^*)$ and $\pi(S)$, $S \in \Xi_{[n]}$. Posterior

$$\pi(\theta^*, S|y) \propto \pi(S)\, \pi(\theta^*|S)\, f(y|\theta^*, S).$$

Alternatively, each y_i has its own θ_i so $y_i \sim f(\cdot|\theta_i),\ i \in [n]$

θ's in a cluster are equal (so for $i \in S_k$, $\theta_i = \theta_k^*$).

Prior for $\theta = (\theta_1, \ldots, \theta_n)$ has atoms as $\Pr(\theta_i = \theta_j) > 0$ so...

(1) use a simulation algorithm to define the NP prior:

$$G \sim \Pi,\ \theta_i \sim G \text{ iid } i \in [n].$$

Unique $\theta' s$ are θ_k^* and $S_k = \{i \in [n] : \theta_i = \theta_k^*\}$.

(2) work out $\pi(S)$ and $\pi(\theta^*|S)$ determined by (1).

If G is a random discrete distribution then the θ's will have ties.

The Dirichlet Distribution - reference slide

Let $w = (w_1, ..., w_M)$ with $\sum_k w_k = 1$.
Suppose $w \sim \text{Dirichlet}(\alpha_1, ..., \alpha_M)$. The density of w is

$$\pi(w_1, w_2, ..., w_M) = \frac{\Gamma(\sum_k \alpha_k)}{\prod_k \Gamma(\alpha_k)} w_1^{\alpha_1 - 1} w_2^{\alpha_2 - 1} ... w_M^{\alpha_M - 1}.$$

Conjugate prior to multinomial: if

$$(n_1, ..., n_M) \sim \text{Multinom}(n, w)$$

with $n = \sum_k n_k$ and

$$w \sim \text{Dirichlet}(\alpha_1, ..., \alpha_M)$$

then

$$\pi(w | n_1, ..., n_M) \propto w_1^{\alpha_1 + n_1 - 1} w_2^{\alpha_2 + n_2 - 1} ... w_M^{\alpha_M + n_M - 1}$$

so

$$w | n_1, ..., n_M \sim \text{Dirichlet}(\alpha_1 + n_1, ..., \alpha_M + n_M).$$

The Multinomial Dirichlet process

We want to distribute probability randomly in Ω. Let $H(d\theta)$ be some simple parameteric distribution on Ω. If $\theta \in R^p$ then

$$H(d\theta) = h(\theta)d\theta$$

for some simple density $h(\theta)$ on R^p. Probability space $(\Omega, \mathcal{B}, H)$.

We can make a very simple random discrete measure as follows.

(!) Fix α and the number of points $M \geq 1$ in the support of G.

1. For $k = 1, ..., M$, sample $\theta_k^* \sim H$.
2. Sample $w_1, ..., w_M \sim$ Dirichlet$(\alpha/M, \ldots, \alpha/M)$.
3. Set $G_M(d\theta) = \sum_{k=1}^{M} w_k \delta_{\theta_k^*}(d\theta)$.

Random probability masses w_k at a random locations θ_k^* in Ω.

Write $G_M \sim \Pi_M(\alpha, H)$ for realisation of process. Say $\theta \sim G_M$.

For $A \in \mathcal{B}$ let $\Pr(\theta \in A | G_M) = G_M(A)$. We have

$$\begin{aligned} G_M(A) &= \int_\Omega \mathbb{I}_{\theta \in A} G_M(d\theta) \\ &= \int_{\Omega \cap A} \sum_{k=1}^{M} w_k \delta_{\theta_k^*}(d\theta) \\ &= \sum_{k=1}^{M} w_k \mathbb{I}_{\theta_k^* \in A}. \qquad (\star) \end{aligned}$$

In distribution notation $G_M = \sum_{k=1}^M w_k \delta_{\theta_k^*}$.

We have a simple procedure that assigns a random probability mass $G_M(A)$ to sets $A \subseteq \Omega$.

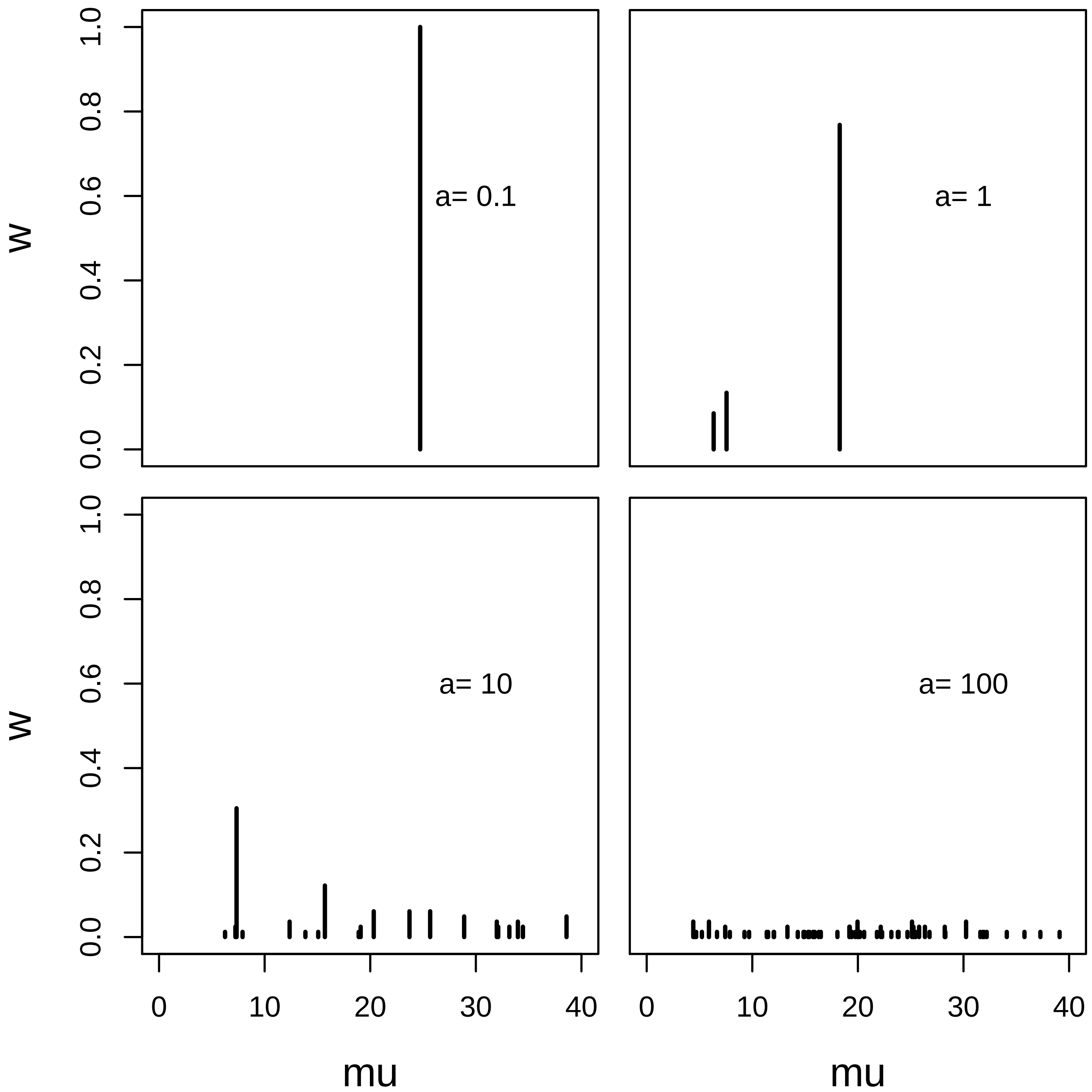
By putting a prior on w and θ^* we determine a prior *on a probability distribution*.

The choice $w \sim$ Dirichlet$(\alpha/M, ..., \alpha/M)$, $\theta_k^* \sim H,\ k = 1, ..., M$ determines a *Multinomial Dirichlet process* $G_M \sim \Pi_M(\alpha, H)$.

G_M is a random distribution, but it is "centred" on the "base distribution" H. From Eqn $(\star)$, since w and θ^* are independent,

$$\begin{aligned} E(G_M(A)) &= \sum_{k=1}^{M} E(w_k) E(\mathbb{I}_{\theta_k^* \in A}) \\ &= \sum_k \frac{\alpha/M}{\sum_j \alpha/M} H(A) \\ &= H(A) \end{aligned}$$

Exercise: use the Dirichlet-variance formula to calculate var$(G_M(A))$.



a= 0.1
a= 1
a= 10
a= 100
w
w
mu
mu
0.0
0.2
0.4
0.6
0.8
1.0
0
10
20
30
40

The Dirichlet process

Definition: $G \sim \Pi(\alpha, H)$ iff $\forall$ partitions $A_1, A_2, ..., A_r$ of Ω

$$G(A_1), ..., G(A_r) \sim \text{Dirichlet}(\alpha H(A_1), ..., \alpha H(A_r))$$

Does such a distribution exist? If it did and $G \sim \Pi$ then $\theta_1, ..., \theta_n \sim G$ (iid) gives an infinite exchangeable sequence.

Conversely, we give an algorithm below realising an IES $\theta_1, ..., \theta_n$ directly given α, H. By de Finetti a random variable G and $d\Pi(G; \alpha, H)$ must exist.

The DP is also the limit $M \to \infty$ of the MDP Π_M (notes appendix and PS4). The definition above "starts afresh".

Properties: we have

$$G(A), G(A^c) \sim \text{Beta}(\alpha H(A), \alpha(1 - H(A)))$$

so $E(G(A)) = H(A)$ as before.

If $G \sim \Pi(\alpha, H)$ and $\theta \sim G$ then marginally $\theta \sim H$, because

$$\Pr(\theta \in A) \;=\; E(E(\mathbb{I}_{\theta \in A}|G)) \;=\; E(G(A)) \;=\; H(A)$$

for all $A \in \mathcal{B}$ so $\theta \sim H$.

If $G \sim \Pi(\alpha, H)$ and $\theta \sim G$ then for $B \in \mathcal{B}$ with $B \subseteq A$,

$$\Pr(\theta \in B|\theta \in A) \;=\; \frac{\Pr(\theta \in B)}{\Pr(\theta \in A)} \;=\; \frac{H(B)}{H(A)}$$

If $H(d\theta) = h(\theta)d\theta$ then marginally $\theta|\theta \in A$ has density

$$p(\theta|\theta \in A) = h(\theta|\theta \in A).$$

DP generative model and predictive distributions

Our generative model for $\theta = (\theta_1, ..., \theta_n)$ is

1. $G \sim \Pi(\alpha, H)$
2. $\theta_i \sim G, i = 1, 2, ..., n$

Here G is a "parameter" like θ. We want to marginalise out G.

Simulate $\theta \sim d\pi(\theta)$ by simulating along the sequence

$$d\pi(\theta) = d\pi(\theta_n|\theta_{1:n-1})d\pi(\theta_{n-1}|\theta_{1:n-2})...d\pi(\theta_1).$$

Suppose $G|\theta_{1:j} \sim \Pi(\tilde{\alpha}, \tilde{H})$. To simulate $\theta_{j+1}|\theta_{1:j}$

1. $G|\theta_{1:j} \sim \Pi(\tilde{\alpha}, \tilde{H})$
2. $\theta_{j+1} \sim G|\theta_{1:j}$.

Find $\tilde{\alpha}$ and $\tilde{H}$ to get a formula for $d\pi(\theta_{j+1}|\theta_{1:j})$.

Claim: The conditional distribution of $G|\theta_1$ is

$$G|\theta_1 \sim \Pi\left(\alpha+1, \frac{\alpha H + \delta_{\theta_1}}{\alpha+1}\right),$$

so if $\theta_2 \sim G|\theta_1$ then marginally $d\pi(\theta_2|\theta_1) = \dfrac{\alpha H(d\theta_2) + \delta_{\theta_1}(d\theta_2)}{\alpha+1}$.

Proof: let $A_1, ..., A_r$ partition Ω. Say $\theta_1 \in A_j$. We have

$$\begin{aligned}\pi(G_1, ..., G_r|\theta_1) &\propto p(\theta_1|G_1, ..., G_r) \times \pi(G_1, ..., G_r) \\ &\propto h(\theta_1|\theta_1 \in A_j)\, G_j \times \prod_{i=1}^{r} G_i^{\alpha H_i - 1}\end{aligned}$$

writing G_j and H_j for $G(A_j)$ and $H(A_j)$. It follows that

$$\begin{aligned}G_1, ..., G_r|\theta_1 &\sim \text{Dirichlet}(\alpha H_1 + \mathbb{I}_{\theta_1 \in A_1}, ..., \alpha H_r + \mathbb{I}_{\theta_1 \in A_r}) \\ &\sim \text{Dirichlet}(\tilde{\alpha}\tilde{H}_1, ..., \tilde{\alpha}\tilde{H}_r)\end{aligned}$$

This matches $\Pi(\tilde{\alpha}, \tilde{H})$ if we choose $\tilde{\alpha}$ and $\tilde{H}$ so that

$$\tilde{\alpha}\tilde{H}(A_j) = \alpha H(A_j) + \mathbb{I}_{\theta_1 \in A_j}.$$

for $j = 1, ..., r$. If we take $\tilde{\alpha} = \alpha + 1$ and base distribution

$$\tilde{H} = \frac{\alpha H + \delta_{\theta_1}}{\alpha + 1}, \qquad \text{(ie } \tilde{H} \text{ normalised)}$$

then this works, so indeed $G|\theta_1 \sim \Pi\left(\alpha + 1, \frac{\alpha H + \delta_{\theta_1}}{\alpha+1}\right)$.

The predictive distribution of $\theta_2|\theta_1 \sim d\pi(\theta_2|\theta_1)$ is

$$\theta_2|\theta_1 \;\sim\; \frac{\alpha H + \delta_{\theta_1}}{\alpha + 1}.$$

$G|\theta_1 \sim \Pi(\tilde{\alpha}_1, \tilde{H}_1)$ and $\theta_2 \sim G|\theta_1$ so marginally $\theta_2|\theta_1 \sim \tilde{H}_1$. [EOP]

Exercise (PS4): if $\theta_1, ..., \theta_j \sim G$ with $G \sim \Pi(\alpha, H)$, show

$$G|\theta_{1:j} \sim \Pi\left(\alpha + j, \frac{\alpha H + \sum_{i=1}^{j} \delta_{\theta_i}}{\alpha + j}\right).$$

Hint: induction replacing G by $G|\theta_{1:j}$.

From the exercise, the predictive distribution of θ_{j+1} is

$$\theta_{j+1}|\theta_{1:j} \;\sim\; \frac{\alpha H + \sum_{i=1}^{j} \delta_{\theta_i}}{\alpha + j}.$$

Sequential simulation: Generative model for $\theta = (\theta_1, ..., \theta_n)$

1. $\theta_1 \sim H$
2. for $j = 1, ..., n-1$
   (a) With probability $\alpha/(\alpha + j)$ simulate $\theta_{j+1} \sim H$.
   (b) Otherwise simulate $\theta_{j+1} \sim U\{\theta_1, ..., \theta_j\}$.

No G in sight!